AKAR KAUNG (He/Him/His)

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EDUCATION

M.Sc. in Computer Science, 2023
University of Minnesota, Twin Cities, MN
GPA: 3.78/4.00
Relevant Coursework: Intelligent Robotic Systems, Parallel Animation & Planning in Games, Computer Organization, Advanced Algorithms and Data Structures, Software Engineering II

B.Sc. in Computer Science, 2021
University of Minnesota, Twin Cities, MN
GPA: 3.59/4.00
Relevant Coursework: Machine Architecture and Organization, Introduction to Operating System, Development of Secure software system, Software Engineer I, Computer Network

RESEARCH EXPERIENCES

University of Minnesota - Twin Cities **May 2022 – Dec 2022**
Research Assistant (Prof. Mattia Fazzini's Lab)

- Collaborated with PhD students to develop tools to streamline data preprocessing and analysis workflows for Natural Language Processing (NLP) models research, enhancing overall efficiency.
- Contributed to an automated project creation system to facilitate course-related assignments and improve reproducibility for Prof. Fazzini's course.
- Conducted bug pattern analysis to predict future software defects, applying data analysis techniques to enhance software quality assurance processes.

Coursework Research Projects

- **Analysis of Cloth Simulation in Parallel Programming** **Jan 2023 – May 2023**
  Prof. Pen-Chung Yew
  Researched and implemented parallel programming techniques to optimize computational efficiency of cloth simulations. Evaluated different approaches to improve performance and achieve smoother real-time rendering of complex physical interactions.

- **Optimized and Efficient Path-Finding Algorithms** **Sept 2022 – Dec 2022**
  Professor Nikolaos Papanikolopoulos
  Conducted comparative analysis of pathfinding algorithms in simulated restaurant environments to identify optimal strategies for robotics navigation, aiding decision-making for operational improvements and customer experience enhancement.

- **Virtual Reality for Social Connection and Well-Being in the Elderly** **Sept 2022 – Dec 2022**
  Professor Victoria Interrante
  Collaborated with Mindvue to research the effects of VR nature interactions on elderly well-being, focusing on immersive environment design and user-centric applications to improve quality of life.

WORK EXPERIENCES

Edirq Inc. **Jan 2025 – Present**
Co-Founder & Software Engineer

- Reduced infrastructure downtime by 30% by designing and deploying a scalable AWS architecture (EC2, RDS/PostgreSQL, SES, SNS) with isolated dev/stage/prod environments, automated backups, and cross-region failover.
- Accelerated deployment by 80% via CI/CD pipelines (GitHub Actions) that included linting, unit testing, staging deploys, and rollback workflows.
- Improved system security and modularity by implementing role-based access, input validation, and fault-tolerant Node.js APIs backed by PostgreSQL.

- Maintained cross-platform API consistency by collaborating with React Native and React teams to define shared data contracts.
- Guided long-term technical direction through engineering roadmaps, design docs, and internal scalability reviews.
- Built a product-market fit (PMF) analytics system to track user signups, session data, feature usage, and DAU/MAU metrics, enabling data-driven growth decisions.
- Evaluated emerging technologies and selected optimal tools to meet business and technical goals.

Meta - Contractor **Mar 2025 – May 2025**
Software QA Tester
- Validated performance, usability, and stability of Meta's AR/VR products by executing 500+ hands-on test cases across Oculus Quest 2, 3, 3S, Meta Ray-Bans, and unreleased prototypes—primarily targeting AI-driven features.
- Identified and documented 50+ high-impact bugs in software, AI behavior, and system performance through functional, exploratory, and usability testing—accelerating issue resolution and improving product quality.

Alum Rock Unified School District **Mar 2024 – Present**
Substitute Teacher
- Provided instructional coverage across 22 schools (TK–Grade 8), ensuring educational continuity in diverse and dynamic classroom environments.
- Adapted teaching strategies and classroom management techniques to align with varying school cultures, curricula, and student needs.
- Delivered prepared lesson plans while maintaining flexibility to support students with different learning styles, ages, and backgrounds.

Arch Gateway - Contractor **Dec 2024 – Mar 2025**
Web Developer
- Oversaw all technical operations for the organization's technology team, including administration of G Suite and internal tools to support streamlined communication and collaboration.
- Designed, developed, and hosted the company website with a focus on education content, performance, usability, and responsive design.
- Implemented SEO strategies to boost visibility and drive organic traffic, resulting in improved search rankings and increased user engagement.

SuperTech FT **Feb 2024 – May 2024**
Software Engineer Internship
- Enhanced the full-stack educational software suite, facilitating real-time communication between a React frontend and a Flask backend deployed on Raspberry Pi.
- Redesigned and maintained the organization's website using JavaScript, HTML, and CSS to ensure modern functionality and aesthetics.

Outlier AI **Jan 2024 – Dec 2024**
AI Prompt Engineer
- Designed and engineered prompt solutions to improve system performance and reliability, yielding an 80% approval rate for outputs flagged as high-quality responses.
- Analyzed prompt behavior and failure cases by reviewing peer submissions, identifying systemic weaknesses, and applying targeted refinements to improve reliability and robustness.

Green River College **May 2018 – Jun 2019**
International Student Ambassador
- Collaborated with cross-functional teams, including student organizations, administrative staff, and faculty members, to ensure comprehensive event planning and seamless coordination of logistics and resources.
- Acted as a liaison between international students and college administration, advocating for student needs and promoting inclusive campus initiatives to enhance the overall student experience.
- Provided guidance and support to international students, facilitating their integration into campus life and promoting cross-cultural understanding and friendship among the student body.

TEACHING EXPERIENCES

University of Minnesota - Twin Cities **Jan 2022 – May 2023**
Graduate Instructor

- Taught core software engineering topics—including Agile methodologies, Git version control, unit testing, and design patterns in C++—to junior and senior undergraduate students.
- Led individual and group troubleshooting sessions to address diverse software issues and bugs, fostering student confidence and technical proficiency.
- Created instructional materials and lab content for over 1,000 students, emphasizing practical application of theoretical concepts.
- Supervised 30+ teaching assistants over multiple semesters, coordinating grading, managing course logistics, and responding to student inquiries.

University of Minnesota - Twin Cities **Sept 2021 – Jan 2022**
Teaching Assistant (Prof. Mattia Fazzini)
- Worked closely with Professor Mattia Fazzini to analyze and refine lab and assignment instructions, contributing to the enhancement of the learning experience
- Led lab discussion sessions to a class of 105 undergraduate students, facilitating their progress and ensuring successful completion of assignments and projects.
- Conducted thorough evaluations of student performance through project assessments, offering constructive feedback to promote continuous improvement and a deeper understanding of course concepts.

LEADERSHIP EXPERIENCES

Green River College - Auburn, WA
CORE Leader/Organizer, Campus Life **May 2018 – Jun 2019**
- Organized and led welcome overnight camping trips for over 300 new international students every quarter.
- Managed logistics, including transportation, food, volunteers, budgeting, and student coordination.
- Interviewed, trained, and supervised volunteers to support event operations and enhance student engagement.

Vice President, Green River Burmese Club **Dec 2017 – Jun 2019**
- Organized cultural events and community-building activities to support Burmese students at Green River College.
- Collaborated with faculty and student organizations to promote cultural awareness and inclusivity.
- Provided leadership and guidance to club members, fostering a supportive and engaged student community.

Event Organizer, Asian Students Union (ASU) **Dec 2018 – Jun 2019**
- Planned and coordinated cultural events to celebrate Asian heritage and foster community engagement.
- Managed event logistics, including venue booking, budgeting, and marketing, to ensure successful execution.
- Collaborated with student organizations and faculty to promote diversity and cultural awareness on campus.

Decoration Department Lead, Lunar New Year Party **Feb 2019**
- Led the decoration team in designing and setting up festive Lunar New Year-themed decorations for the event.
- Managed volunteers, budget, and materials to create an immersive and culturally authentic atmosphere.
- Collaborated with the college art department to design and craft DIY decorations.

Security Department Lead, Halloween Party **Dec 2018**
- Led the volunteer security team to ensure student safety and smooth event operations during the Halloween Party.
- Collaborated with college security officers to handle potential threats and enforce safety protocols.
- Coordinated crowd management and assisted attendees while addressing minor security concerns.

Web Master, International Student Events (ISE) **Jun 2017 – May 2019**
- Designed, developed, and maintained the official website for International Student Events (ISE) to enhance event visibility and engagement.
- Managed website content, ensuring up-to-date information on events, resources, and student activities.

Gusto College, Yangon, Myanmar **2016**
President - Student Government Board
- Led the Student Government Board in representing student interests and organizing campus initiatives.
- Coordinated events, meetings, and projects to enhance student engagement and academic support.
- Acted as a liaison between students and faculty, advocating for student needs and campus improvements.

VOLUNTEER AND COMMUNITY ENGAGEMENT

Panelist Speaker, Education USA **Jul 2019**
Yangon, Myanmar

- Provided prospective college students with insights on cultural adaptation, time management, and academic success while studying abroad.
- Advised students on what to expect in a new educational system and strategies for a smooth transition to international studies.

Volunteer, City Forest Clean Up **May 2018**
Auburn, WA

- Supported environmental preservation efforts by cleaning up litter and maintaining the health of local forests and green spaces.

Volunteer, Phi Theta Kappa Honor Society **Mar 2017 – Dec 2017**
Green River College, Auburn, WA

- Took part in a community service initiative to remove graffiti and revitalize several neighborhoods.

Volunteer, GRC Welcome Week **Feb 2017**
Green River College, Auburn, WA

- Helped new students transition to college life by leading campus tours, organizing activities, and facilitating engagement through interactive games.

Volunteer, GRC CORE Trip **Aug 2017**
Green River College, Auburn, WA

- Led student groups during an overnight camping trip, facilitated team-building activities and helped new students adjust through storytelling and mentorship.

Volunteer, Water Aid Donation **Mar 2014**
Tayat, Myanmar

- Assisted in distributing clean water supplies to communities affected by drought and helped raise awareness about water scarcity and conservation efforts.

HONORS, AWARDS, AND ACHIEVEMENTS

1. **Dean's List of Scholars, College of Science and Engineering, University of Minnesota, January 2020 – May 2021.** Awarded to students whose grade point average is 3.66 or above and who completed 12 hours of graded coursework per semester.
2. **Campus Life Leadership Award, Campus Life, Green River College, June 2019.** Recognized for outstanding leadership and contributions as an International Student Ambassador, fostering community engagement and supporting fellow students.
3. **Honorable Mention for Best Decoration Event, The Current, Green River College, June 2019.** Recognized for creating a vibrant and culturally authentic atmosphere, featuring traditional lanterns, calligraphy, and symbolic decorations that captured the essence of the Lunar New Year celebration.
4. **Student Representative of Math Division Tenure Committee, Green River College, January 2019 - June 2019.** Served as one of the few selected student representatives on the Math Division Tenure Committee, contributing to faculty tenure evaluations and decision-making.
5. **Student Representative of IT Budget Committee, Green River College, May 2018 - June 2019.** Served as the sole student representative on the IT Budget Committee, providing input on technology funding decisions and advocating for student needs in IT resource allocation.
6. **Awarded to participate in Council of Unions and Student Programs - Student Leadership Conference, Green River College, May 2018.** Awarded the opportunity as a selected student from Green River College to participate in the Council of Unions and Student Programs - Student Leadership Conference, representing the college in leadership development and advocacy discussions.
7. **Certificate of Recognition for Thesis Defense, American Mathematical Association of Two-Year Colleges, Green River College, June 2018.** Recognized for successfully presenting a thesis defense in the 2018 AMATYC Student Research League, an exclusive competition for selected participants chosen by the Green River College Mathematics Division.

8. **Recognition of Outstanding Achievement in Mathematics, Green River College, June 2018.** Awarded MAA membership by the Green River Mathematics Division for exceptional academic performance and excellence in mathematics.

9. **Best Design Award ChatBot Hackathon, Myanmar, November 2016**. Awarded $750 for developing a chatbot with optimized UI/UX and multilingual support, enhancing user engagement and efficiency.

Projects

Avis Budget Group Commission Payout Calculator (URL)
Tech Stack: Android Studio, Java, Gradle, XML, Figma

- Developed a commission payout calculator for agents to use at the Seattle location to calculate their earnings quickly and easily using their mobile devices.
- Designed a user-friendly interface, ensuring easy navigation and data entry, resulting in a more intuitive and efficient tool.
- Attained ultimate user satisfaction through effective development and proactive issue resolution for the application.

Aurora ET (URL)
Tech Stack: Android Studio, Java, Gradle, XML, Node JS, MongoDB, Figma

- Developed an Android application to track aurora activity using real-time data from the National Oceanic and Atmospheric Administration Government Agency KP index, providing users with timely alerts and interactive visualizations.
- Designed a user-friendly interface, ensuring easy navigation and data entry, resulting in a more intuitive and efficient tool.
- Incorporated efficient backend systems with Node.js and MongoDB for real-time data processing and storage.

Space-Warfare (URL)
Tech Stack: UE5, C++, Blender

- Created smart spaceship simulation in UE5, predicting and engaging enemy spaceships while implementing evasion tactics.
- Designed a self-destruct feature for the spaceship, triggered when health drops below a threshold.
- Integrated advanced evasion tactics into the spaceship's AI algorithms, enabling it to successfully dodge and outmaneuver enemy spaceships when detected and targeted.

Back2Town (URL)
Tech Stack: Unity, C#, Blender

- Developed a 3D character simulation within Unity, incorporating advanced NPC movement using the Dijkstra pathfinding algorithm.
- Designed and implemented a sophisticated collision avoidance system to enhance gameplay realism and character interactions within the virtual environment.
- Utilized Unity's tools to bring detailed, realistic environments to life, improving both visual appeal and functionality.

Darkdown (URL)
Tech Stack: HTML, CSS, Tailwind, Javascript, Vue, SQL, Agile, Firebase, GoogleAuth

- Developed an advanced note-taking feature catering to coders, integrating live Markdown syntax and shortcuts.
- Engineered device integration capabilities by allowing users to capture and upload photos directly from their mobile devices.
- Integrated a responsive application for desktop and mobile use, offering the flexibility of app download for an enhanced user experience.
- Designed and structured the database architecture using Firebase's NoSQL database, employing collections, documents, and subcollections to efficiently organize user-centric data entities.

Dribbbl (URL)
Tech Stack: Python, Flask, HTML, CSS, Javascript, PostgreSQL, Auth0, Agile, Jinja

- Created a full-stack web application using Flask, implementing secure user authentication and data storage with PostgreSQL.
- Built a user authentication system using Flask, Authlib, and Auth0.
- Developed RESTful APIs for managing user data with optimized data flow between the front-end and back-end systems.

Autonomous Drone Transportation Simulation (URL)
Tech Stack: C++, HTML, CSS, Javascript, CSS, UML, Agile

- Developed a simulation for autonomous passenger drones, including algorithmic control of drone flight and passenger transport functionality.
- Produced detailed UML diagrams and comprehensive documentation, supporting code optimization and system architecture clarity.
- Refined existing codebase using Factory, Decorator, and Strategy design patterns to improve code organization and structure.

US Voting System (URL)
Tech Stack: Java, JUnit, Shell, Sprint

- Developed the architecture and system design to import ballots and generate election results, adhering to various voting rules/methods specified by the United States Parliament Voting System.
- Created comprehensive testing cases using JUnit to ensure a thorough examination of software functionality and robust code quality.
- Documented a Software Requirements Specification, delineating the application's specifications and functional requirements, including a thorough comparison of the required hardware for optimal software utilization.

Telekinesis VR
Tech Stack: Unity, C#

- Pioneered the concept of telekinesis interaction and brought it to the forefront of discussion.
- Developed the algorithm for manipulating objects in 3D space.
- Conducted a thorough analysis and optimization of the interaction between the user and object to ensure a smooth user experience.

Gopher Chat Room (URL)
Tech Stack: C, TCP, IPv4, Threads

- Designed and developed a virtual chat room where users on different hosts can join and send messages.
- Implemented features of communicating between different users by multi-threading, multiprocessing communication, TCP, and IPv4.

Directory Locker (URL)
Tech Stack: Java, RSA, AES-GCM, Java NIO, Java Security API, SecureRandom, Base64

- Developed and implemented cryptographic functionalities using RSA and AES-GCM algorithms, ensuring secure data transmission and storage in a custom Java application.
- Applied Java Security API for robust cryptography practices, integrating SecureRandom for secure key generation and Base64 encoding to ensure data integrity and confidentiality.
- Conducted performance optimizations and security testing to validate encryption robustness and minimize data leakage vulnerabilities in secure communication scenarios.

Cloth And Fluid Simulation (URL)
Tech Stack: Processing

- Developed a cloth simulation and fluid dynamics model using processing to simulate realistic cloth movement and interactions with fluid, achieving smoother simulations.
- Integrated collision detection to ensure realistic interactions between the cloth and surrounding objects, improving the overall realism of the simulation.
- Designed and integrated a computational model for real-time visualization, demonstrating realistic cloth behavior and fluid interactions in dynamic environments.

Core NLP XML To Brat Ann Conversion (URL)
Tech Stack: Python, XML, ANN

- Developed a Python script to convert Core NLP XML files to Brat Ann format, handling Part-of-Speech (POS) tagging and dependency relations based on Enhanced++ Dependencies.
- Implemented functions for parsing XML structures, extracting token information, and generating annotation and configuration files, ensuring compatibility with Brat visualization tools.
- Utilized XML parsing techniques in Python’s ElementTree library to automate the conversion process, improving efficiency and reducing manual annotation work.